BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. GPU.

Q2. Actually running independent work at same time.

Q3. A task needs another task's result.

Q4. 12.

### You Do pages · Independent questions

Q1. Many parallel lanes/threads [2], pixels independent [2], throughput on similar operations [2].

Q2. AI tensor/matrix workloads [3], scientific modelling independent calculations [3] or valid alternatives.

Q3. Short sequential: CPU avoids transfer/parallel overhead [2], massive independent: GPU throughput [2], branches/memory/bandwidth caveats [2].

### Exit Ticket · Model answer

False. GPUs also accelerate AI and scientific computation.

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

Do Now Recall previous learning | I Do Read new knowledge | We Do Practise with guidance | You Do Work independently | Review & Improve Check my understanding



RHS COMPUTING / OCR H446 / mixed LESSON 100 · DRAFT

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

# LESSON 100 · Lesson 100 · GPUs: Parallel Work Beyond Graphics

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. Which processor is designed for throughput on massive parallel workloads?

Q2. What is parallelism?

Q3. What is a data dependency?

Q4. An image array has 4 rows × 3 columns. How many pixels?

Do Now Recall previous learning | I Do Read new knowledge | We Do Practise with guidance | You Do Work independently | Review & Improve Check my understanding

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

GPU — Graphics Processing Unit.

SIMT — Single instruction, multiple threads execution model.

throughput — Total work per time.

divergence — Threads follow differing control paths.

## New learning

Observe — GPUs provide many execution units designed for high-throughput parallel operations on large batches of data.

Apply — Besides graphics, GPUs can accelerate machine learning, physical simulations, image processing and some scientific calculations.

Evaluate — Transfers between memory/device, limited parallelism and divergent branching can outweigh GPU speedup on small or sequential tasks.

## Worked example

Explain why processing one million independent pixels may suit a GPU.

1. Follow the reasoning — Each pixel can receive similar operation in parallel; many GPU threads process different positions.

# We Do · Practise with guidance

A sequential data-dependent algorithm calculates next item using previous result. Does GPU automatically accelerate it?

Follow the reasoning

Not necessarily: dependence limits simultaneous work, and data-transfer overhead may dominate.





**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

## Question 1

Explain two features that make a GPU useful for applying same filter to millions of pixels. [6]

## Question 2

Give two non-graphics GPU applications and explain how parallelism helps. [6]

## Question 3

Compare CPU and GPU for 12-item sequential task versus million-element independent matrix operations. [6]

# Exit Ticket · Answer before checking

A GPU is used exclusively for drawing computer-game graphics.